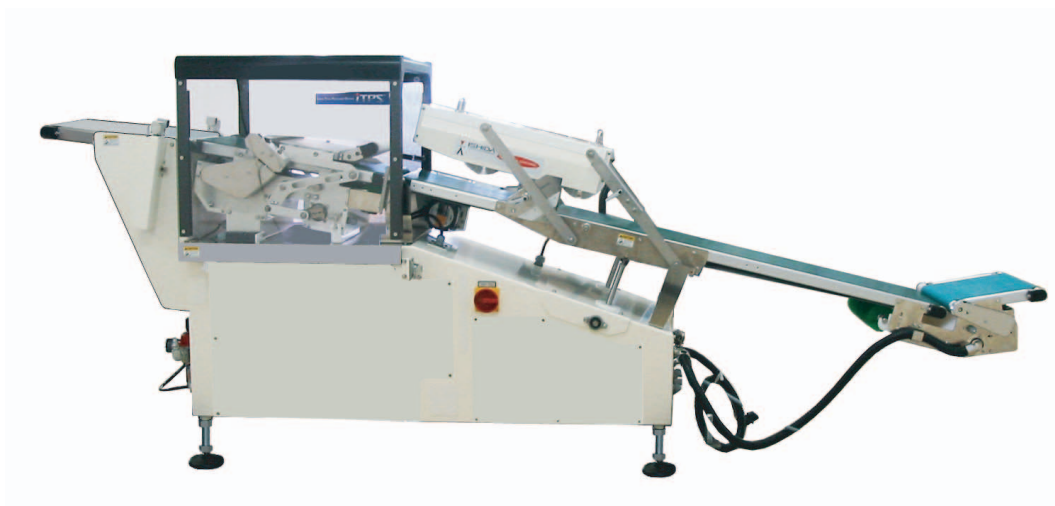




THICKNESS/SEAL CHECKER

TSC-RS-120

MAINTENANCE MANUAL



WARNING

- Do not carry out installation, operation, service, or maintenance until thoroughly understanding the contents of this manual.
- Keep this manual available at all times for installation, operation, service, and maintenance.

ISHIDA CO., LTD.

No. 085-8863-061
Published in November 2008

You can help improve this manual by calling attention to errors and by recommending improvements.
Please convey your comments to the nearest Ishida Company regional representative.

Thank you!

Copyright (c) 2005 by Ishida Co.,Ltd. All Rights Reserved.

No part of this manual may be reproduced in any form, by photocopy or any other means,
without permission in writing from the publisher.

PREFACE

Purpose

This service manual is designed to help service personnel maintaining TSC-RS-120. Start the actual maintenance after reading through this manual.

Refer to the Operation Manual about installation, detailed setting and operations of the machine.

Related Manuals

- TSC-RS-120 Operation Manual
- TSC-RS-120 Parts List

Graphical Labels

The following graphical labels are used in this manual:

Graphical	LabelMeaning
	Indicates a potentially hazardous situation which, if not avoided, could possibly result in death or serious injuries. (The label is used to indicate a potentially hazardous situation which, if not observed, could possibly result in death or serious injuries.)
	Indicates a potentially hazardous situation which, if not avoided, could possibly result in minor or moderate injuries or machine damage. (The label is used to indicate a potentially hazardous situation, if not observed, could possibly result in minor injuries.)
	Indicates special cautions or important information.
	Indicates helpful information.

For Safety Service

Strictly observe the following items to perform proper maintenance and prevent accidents from occurring.

- Regarding assembly and adjustment sections, precautions are mentioned for each item. Thoroughly understand them to perform proper work.
- Always keep the surroundings of the equipment organized. In particular, extreme caution is must be taken during disassembly, because serious damage may occur if the power switch is turned ON with screws left in the equipment.
- Turn OFF the power switch before starting maintenance. According to circumstances, start maintenance after removing the power supply cable from the power receptacle.

IMPORTANT CONSIDERATIONS



WARNING

It is imperative to read and understand the contents of this manual thoroughly when operating the machine or performing any service, maintenance, or inspection procedures. Please be aware of the hazards involved which may not be obvious, and follow the instructions detailed in this manual.

1. It is impossible to enumerate or predict all of the possible dangers of using this equipment and fully list all of them in an instruction manual. When operating the equipment or performing production in any way not specifically described in this manual, please contact the distributor where the equipment was purchased or your Ishida customer service representative before proceeding.
Safety countermeasures not specifically described in this manual or indicated on the machine itself should be carefully considered and implemented before performing any operation, service, or maintenance procedures.
2. The copyright for the material in this document is held by Ishida Co., Ltd., and all information contained herein is protected by the Copyright Law. Any disclosure to third parties or unauthorized copying of the information contained herein is not permitted without prior written approval from Ishida Co., Ltd.
3. Although this manual has been carefully edited, if there are any mistakes, or if you have any questions, please contact your local Ishida Service representative or its local distributor/representative.
The model number and machine number are printed on the equipment plate attached to the equipment.
4. The information contained herein may be changed without prior notice.

 ISHIDA			
MODEL _____			
PRODUCT _____			
WEIGHING RANGE		g~	g
JOB NUMBER _____			
NUMBER _____			
MFG DATE _____			
POWER SINGLE	AC	V	A
3-PHASE	AC	V	A
FREQUENCY		Hz	
ISHIDA CO.,LTD. MADE IN JAPAN			

Equipment Plate

WARRANTY CONDITIONS

1. Ishida Co., Ltd. only assumes responsibility for repairing or replacing components which are deemed to be defective due to improper design or manufacturing.
The measures to be taken for defects whose origin is unclear shall be decided by mutual consultation between both parties.
2. Ishida assumes no responsibility for loss, injury or damage which are the result of unauthorized or unforeseen operational procedures.
3. The warranty period is stated in the Certificate.

Note

- The definition for each intended reader described in the table below is as follows:

- | | |
|----------------------------------|--|
| • Operator: | Personnel who perform basic operations during daily production with the production line (i.e. Operator Level).
Do not perform any work described in this manual if it is targeted to any personnel other than Operator. |
| • System manager: | Personnel who set weighing parameters or tune the equipment (i.e. Site Engineer Level).
Do not perform any work described in this manual if it is targeted to Maintenance personnel. |
| • Maintenance personnel: | Personnel who have specialized knowledge about the equipment and perform maintenance or inspection of the equipment (i.e. Installation Engineer Level).
Do not perform any work described in this manual if it is targeted to Service personnel. |
| • Ishida Service representative: | Personnel who perform tuning when installing the equipment.
There is no limitation in performing works described in this manual unless specially. |

Table of Contents

GENERAL TABLE OF CONTENTS

1 CONFIGURATION OF THE SYSTEM

1.1	Specification	1-1
1.2	Configuration.....	1-2
1.3	Device Location.....	1-4

2 REPLACEMENT AND ADJUSTMENT

2.1	Sensor Replacement.....	2-1
2.1.1	Position of each Sensor / Motor in Infeed Conveyor Height ADJ. Unit.....	2-1
2.1.2	Seal check Conveyor Sensor	2-2
2.2	DC Motors	2-3
2.2.1	DC Motor Replacement	2-3
2.2.2	Reject Unit	2-5
2.2.3	Bag Stabilizer Unit (Option)	2-6
2.2.3.1	Position of each Sensor / Motor in New Stabilizer Unit.....	2-6
2.2.4	Operation (HBL Series DC Motor)	2-7
2.2.4.1	Operating the motor	2-7
2.2.4.2	Operating conditions	2-7
2.2.4.3	Example of operational timing chart	2-8
2.2.4.4	Protection functions.....	2-8
2.2.4.5	Troubleshooting	2-9
2.2.5	Setting the running speed (AXH Series DC Motor)	2-11
2.2.6	Motor rotation direction setting	2-15
2.2.7	Emergency switch.....	2-16
2.3	AC Servo Motor.....	2-18
2.3.1	AC Servo Motor Replacement	2-18
2.3.2	AC Servo Parameter Instruction Guide	2-19
2.4	Timing Belt Tension Adjustment.....	2-26
2.5	Conveyor Belt Tension Adjustment	2-27
2.6	Test method of seal check conveyor alignment	2-28

3 ELECTRICAL CONTROL UNIT

3.1	Locations of Electrical Devices	3-1
3.1.1	P.S UNIT	3-1
3.1.2	ADJUSTMENT P.S UNIT	3-3
3.1.3	CONTROL UNIT	3-5
3.1.4	TOTAL DIAGRAM	3-7
3.2	SSC CONTROL BOARD (P-5530*)	3-8
3.3	DMU BOARD (P-5562*)	3-22
3.4	Extended I/O BOARD (P-5500*)	3-29
3.5	Control Panel Items (Maintenance Service) (Option)	3-33
3.5.1	Display Control Menu Screens	3-33
3.5.1.1	Touch Panel Coordinate Adjustment	3-33
3.5.2	Password Set/Language Select Set Menu Screen	3-35
3.5.2.1	Language Select Setting	3-35
3.5.2.2	Password Setting	3-36
3.5.3	Destination ID Menu Screen	3-37
3.5.3.1	Destination ID	3-37
3.5.3.2	Browser Setting	3-37
3.5.3.3	E-mail Setting	3-37
3.5.4	Communication Set Menu Screen	3-38
3.5.4.1	[RCU] Communication Setting	3-38
3.5.4.2	[Main Body] Communication Setting	3-38
3.5.4.3	[Server IP Address] Setting	3-39

4 EXPERT SETTINGS

4.1	Machine Parameter screen	4-1
4.1.1	Drive Unit Setting screen	4-2
4.1.2	PID Control Setting Screen	4-3
4.1.3	Motor Setting Screen	4-4
4.1.4	System Diagnosis screen	4-5
4.1.4.1	System Diagnosis screen 1 (GPIO)	4-6
4.1.4.2	System Diagnosis screen 3 (BL-MOTOR)	4-8
4.1.4.3	System Diagnosis screen 6 (OTHER)	4-9
4.1.5	Program Number Screen	4-10
4.1.6	Memory Inti. & Copy Screen	4-10

5 APPENDIX

5.1	Program Installation Procedure from Web-RCU	5-1
5.1.1	Installation Software Start up	5-1
5.1.2	RCU Software Backup	5-2
5.1.3	RCU Software Installation	5-4
5.1.3.1	Installing the Program to Existing Model	5-4
5.1.3.2	Installing Program Newly	5-5

6 TROUBLESHOOTING

6.1	Error Message Handling	6-1
6.2	System Error / Failure	6-3
6.3	Other Troubleshooting	6-3
6.4	TSC-RS-120 Troubleshooting (in operation)	6-4
6.5	Error detect and checkup	6-6

1 CONFIGURATION OF THE SYSTEM

1.1	Specification	1-1
1.2	Configuration	1-2
1.3	Device Location	1-4

<MEMO>